

SWIFT CURRENT CDA, SK, Canada

WMO: 714460

Lat: 50.260N

Lon: 107.737W

Elev: 825

StdP: 91.80

Time Zone: -6.00 (W06)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -30.0	(c) -26.9	(d) -33.3	(e) 0.2	(f) -30.0	(g) -29.9	(h) 0.3	(i) -26.8	(j) 15.8	(k) -4.8	(l) 14.0	(m) -6.7	(n) 5.0	(o) 280	(p) 0.777

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.6	31.2	17.5	29.1	17.0	26.9	16.3	19.9	26.8	18.6	25.5	17.4	24.4	5.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	14.0	22.7	16.2	12.8	21.1	15.0	11.7	20.1	60.8	26.8	56.2	25.3	52.3	24.3	25.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 13.1	(b) 11.6	(c) 10.5	(e) -33.5	(f) 34.5	(g) 2.8	(h) 2.2	(i) -35.6	(j) 36.0	(k) -37.2	(l) 37.3	(m) -38.8	(n) 38.5	(o) -40.9	(p) 40.1		
			DB	WB	-33.7	22.0	2.8	1.7	-35.7	23.2	-37.3	24.2	-38.9	25.2	-40.9	26.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.2	-10.4	-9.3	-3.5	4.5	10.5	15.1	18.7	18.1	12.8	5.2	-2.9	-9.1
	DBStd	12.20	8.78	7.92	8.08	5.64	4.76	3.59	3.29	3.86	5.03	5.51	7.28	8.02
	HDD10.0	2996	633	540	418	178	53	4	0	1	27	164	387	592
	HDD18.3	5269	892	773	676	417	245	109	36	53	177	406	636	850
	CDD10.0	887	0	0	1	12	69	156	269	252	111	16	2	0
	CDD18.3	118	0	0	0	0	3	11	47	46	11	0	0	0
	CDH23.3	1527	0	0	0	5	63	151	542	587	173	5	0	0
	CDH26.7	506	0	0	0	1	10	38	181	221	56	0	0	0
Wind	WSAvg	5.6	6.2	5.7	5.9	5.9	5.8	5.4	4.8	4.8	5.2	5.7	5.8	6.0
Precipitation	PrecAvg	389	22	18	21	27	47	70	48	42	36	19	16	23
	PrecMax	573	50	41	71	89	133	222	109	135	99	69	45	56
	PrecMin	265	3	4	3	4	7	20	7	4	3	1	3	9
	PrecStd	79	12	10	17	19	31	44	28	31	27	15	11	13
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.8	7.4	15.5	23.1	28.0	30.5	33.7	34.1	31.3	23.9	16.3	8.0
		MCWB	3.5	3.7	8.3	11.4	14.1	17.6	18.6	17.6	15.7	12.9	8.6	3.2
	2%	DB	4.3	4.5	12.0	19.4	25.1	27.3	30.8	31.6	28.2	19.5	11.5	5.1
		MCWB	1.8	1.9	6.2	9.8	13.6	16.5	18.9	17.1	14.8	10.8	6.3	2.3
	5%	DB	2.9	2.7	9.0	16.5	22.5	24.7	28.6	29.2	24.8	16.1	8.6	3.2
		MCWB	1.0	0.8	4.7	8.2	11.9	15.2	18.5	16.3	13.8	9.2	4.4	1.0
	10%	DB	1.2	1.0	6.1	13.7	19.7	22.6	26.3	26.7	21.6	13.3	5.8	1.3
		MCWB	-0.2	-0.4	3.0	6.9	10.9	14.3	17.4	16.0	12.5	7.8	2.7	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.8	4.1	8.8	12.3	15.9	20.7	22.9	20.2	17.0	13.4	9.0	4.3
		MCDB	5.8	6.8	14.5	20.2	24.3	26.8	28.4	28.2	26.0	22.2	14.9	6.9
	2%	WB	2.1	2.2	6.7	10.5	14.2	17.7	20.6	18.8	15.7	11.4	6.6	2.4
		MCDB	4.0	4.0	11.3	18.1	22.8	23.6	27.0	26.7	25.3	18.2	10.9	4.7
	5%	WB	1.1	1.0	4.9	8.9	12.9	16.4	19.3	17.7	14.4	9.6	4.7	1.1
		MCDB	2.9	2.5	8.7	15.6	20.9	22.5	25.9	25.8	23.1	15.1	8.3	3.2
	10%	WB	-0.3	-0.5	3.0	7.3	11.6	15.2	18.2	16.8	13.1	8.1	2.9	-0.5
		MCDB	1.2	1.0	6.0	13.0	18.3	20.8	24.5	24.6	20.6	12.8	5.5	1.3
Mean Daily Temperature Range	5% DB	MDBR	8.8	8.8	9.4	11.4	12.3	11.0	12.6	13.2	12.4	10.7	8.8	8.3
		MCDBR	9.4	9.5	12.8	16.0	16.3	14.7	15.9	17.4	17.1	15.1	12.0	9.5
		MCWBR	7.8	7.5	8.1	8.5	7.6	6.7	7.0	6.7	7.5	8.3	7.8	7.3
	5% WB	MCDBR	9.3	9.3	12.3	14.7	14.4	12.6	13.7	14.5	15.3	14.0	11.7	9.4
		MCWBR	7.8	7.5	8.1	8.3	7.1	6.6	7.0	6.5	7.2	8.1	7.7	7.4
Clear-Sky Solar Irradiance	taub	0.245	0.246	0.276	0.344	0.364	0.358	0.364	0.384	0.333	0.286	0.256	0.234	
	taud	2.420	2.488	2.502	2.318	2.306	2.384	2.380	2.288	2.405	2.527	2.499	2.444	
	Ebn at Noon	827	926	945	899	892	899	886	847	862	857	809	792	
	Edh at Noon	59	71	85	116	123	115	113	118	94	69	55	49	
All-Sky Solar Radiation	RadAvg	1.22	2.22	3.55	4.84	5.82	6.16	6.68	5.39	3.94	2.42	1.32	0.94	
	RadStd	0.13	0.17	0.26	0.34	0.41	0.42	0.41	0.41	0.35	0.28	0.15	0.09	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	-1.30	-1.69	N/A	N/A	N/A
(47) Regional (16 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page